

Introduction

Air transportation is arguably the most critical transport infrastructure that has been developed over the last century. According to Airports Council International, in 2024, an estimated average of 12.8 million passengers were transported each day globally. However, this unprecedented mobility has also turned the aviation network into a major channel for the global propagation of infectious diseases. The vast heterogeneity and complex architecture of the worldwide airport network generate seemingly erratic and highly complicated epidemic patterns, as documented during the SARS and COVID-19 outbreaks (Colizza et al., 2006). Consequently, traditional homogeneous mixing models are insufficient to predict global pandemics, requiring a systematic investigation based on complex networks theory.

Over time, the aviation network has largely evolved into a "hub-and-spoke" topology, where a minority of airports process the vast majority of traffic. This structure creates a "vulnerability paradox": while the network is highly resilient to random failures, it is extremely vulnerable to targeted attacks on its most connected nodes (Zamin & Lillo, 2013). In epidemiological terms, a targeted "attack" equates to a public health intervention, such as immunization. Research indicates that isolating or vaccinating just a small fraction of the most critical cities is drastically more effective in containing an epidemic than random interventions (Hufnagel et al., 2004). Random vaccination proves highly inefficient in scale-free networks, often requiring the immunization of almost the entire population to halt the spread (Schneider et al., 2011).

Therefore, identifying "super-spreaders" or critical nodes through network centrality measures is vital for optimal vaccine allocation. The most common targeted strategies focus on Degree Centrality (nodes with the highest number of traffic connections) and Betweenness Centrality (nodes that act as shortest-path bridges between different regions). The relative effectiveness of these two metrics is a subject of debate in the literature, presenting two contrasting network models. On one hand, research on metapopulation epidemic models demonstrated that degree-based and betweenness-based interventions can yield nearly identical results in certain systems (Tanaka et al., 2014). This occurs in highly integrated networks where the metrics are strongly correlated, meaning the busiest hubs naturally serve as the primary bridges.

On the other hand, a global topological analysis discovered the existence of an "anomalous centrality" in worldwide aviation (Guimerà et al., 2005). Due to geopolitical constraints and the formation of regional communities, the most connected airports are not necessarily the most central. The authors proved that certain airports with very low degree (low traffic) can exhibit exceptionally high betweenness centrality because they act as unique, indispensable bridges connecting otherwise isolated regions—such as Anchorage serving as the main bridge between the continental United States and the isolated Alaskan community (Guimerà et al., 2005).

This topological paradox leads to a critical gap regarding the "status quo" of the European air transport network. The European airspace is a highly dense and heterogeneous system that mixes major traditional hub-and-spoke operations with extensive point-to-point connections driven by low-cost carriers. Furthermore, during crisis events such as the COVID-19 pandemic, the European network demonstrated a strong tendency to fragment into localized, national communities (Maccurri et al., 2022). It remains an open question whether the current topology of the European network aligns more with the highly correlated, integrated model observed by (Tanaka et al., 2014), the fragmented, anomaly-driven community structure described by (Guimerà et al., 2005), or a unique hybrid of both.

Furthermore, the effectiveness of a targeted vaccination strategy is not solely dictated by network topology; it is fundamentally intertwined with the intrinsic biological properties of the virus. Different diseases—characterized by distinct incubation periods and transmission rates—can be modeled through various compartmental frameworks. A node's capacity to amplify or attenuate an epidemic depends on the local basic reproduction number, R_0 (Lee et al., 2013). Thus, the identification of optimal vaccination targets must adapt to the specific infection dynamics, as a virus with a long latency period may exploit the network's bridge nodes differently than a rapidly contagious pathogen.

In this work, the aim is to study the topology of the European flight network to decipher its current structural status quo, simulating how a disease would spread via the aviation route in Europe and how vaccination can prevent the infection spread by testing different vaccination strategies across different infection models. Different diseases can be modelled through existing infection models (e.g., SIS, SIR, SEIR, SQR) that are defined by different parameters. This way, it is possible to correlate the type of infection with the way it spreads in the network and how vaccination strategies work differently according to the infection type, reinforcing the notion that one immunisation strategy does not work for all epidemics and it should be modelled according to the infection information available.

Vaccination strategies can vary greatly depending on the type of infection currently in study. Random vaccination will target random cities in the network, while vaccination according to different metrics, such as betweenness centrality or degree centrality, will demonstrate an alteration in vaccination efficiency. In this context, the betweenness centrality of the network would represent cities that act as bridges connecting different regions, while degree centrality targets the heaviest density zones. By testing these different vaccination strategies on different infection models, it is possible to effectively interpret how vaccinations change the speed and spread of the infection across the aviation network. Ultimately, evaluating whether degree or betweenness centrality achieves better containment will reveal if the European network behaves as a fully integrated system or a community-based structure, demonstrating that the optimal vaccination strategy must be tailored to both the topological reality of the continent and the biological characteristics of the infection.

Versao atual:

1. Enquadrar o problema: as infraestruturas de transporte e a mobilidade humana influenciam diretamente a propagação espacial de epidemias. Num continente altamente conectado como a Europa, surtos locais podem tornar-se regionais ou continentais através das redes de mobilidade, embora esse processo dependa das características biológicas do agente patogénico.
2. Introduzir os modelos compartimentais clássicos usados no trabalho: SIR, SEIR, SEIQRD, SIS e SEIRS. Explicar brevemente que cada modelo representa um arquétipo epidemiológico diferente, incluindo imunidade permanente, período latente, quarentena, mortalidade, ausência de imunidade ou perda de imunidade.
3. Justificar a abordagem metapopulacional: modelos homogéneos descrevem bem a dinâmica interna de uma população, mas não capturam propagação entre cidades. Assim, cada cidade é tratada como uma população local ligada a outras através de fluxos de mobilidade.
4. Apresentar a Europa como uma rede multilayer de transporte composta por air, water e land layers. Explicar a função epidemiológica de cada camada: aviação para ligações rápidas e longas, ferries para ligações costeiras/insulares, e mobilidade terrestre para commuting e propagação regional.
5. Introduzir a distinção entre mobilidade difusiva e recorrente. Air e water representam deslocações mais difusivas, capazes de introduzir infeção em novos locais. Land/commuting representa mobilidade recorrente, onde indivíduos viajam e regressam, podendo sincronizar epidemias entre cidades próximas. (nao desenvolver demasiado pois ja e coberta de forma extensa na metodologia)
6. Explicar que a topologia da rede influencia a propagação. Algumas cidades ou ligações funcionam como hubs, pontes ou núcleos estruturais, e por isso podem ter impacto desproporcional na difusão epidémica.
7. Introduzir as métricas de ciência de redes usadas para intervenção: degree, betweenness, k-core e motifs. Estas métricas permitem selecionar cidades ou ligações estratégicas para tentar conter a epidemia.

8. Apresentar as estratégias de controlo: intervenções em nós, interpretadas como vacinação/proteção de cidades, e intervenções em arestas, interpretadas como fecho ou restrição de rotas.
9. Clarificar que o objetivo é comparativo e não preditivo. O trabalho não pretende simular uma epidemia real específica, mas estudar como diferentes doenças, camadas de mobilidade e estratégias de intervenção alteram os padrões de propagação.
10. Terminar com a pergunta/objetivo central: perceber como a estrutura multilayer das infraestruturas europeias permite ou limita a propagação de diferentes epidemias, e como métricas de ciência de redes podem ajudar a identificar estratégias eficazes de contenção.

Methodology

Following the work objectives, to evaluate the dynamics of disease propagation, airborne transmission was initially simulated based on air transmission using an aviation network model. For this network, four epidemic models were implemented and evaluated (Susceptible-Infectious-Recovered (SIR), Susceptible-Infectious-Susceptible (SIS), Susceptible-Exposed-Infectious-Recovered (SEIR), and Susceptible-Quarantined-Infectious-Recovered (SQIR). The four simulations were parameterized across varied vaccination strategies, immunization coverage levels and initial patient-zero seeding configurations.

To simulate a more realistic epidemic scenario, additional transmission layers representing land and water routes were integrated, transforming the system into a multilayer network.

For simplification purposes, only European countries were selected from this network, so as to model how an infection would spread across Europe. It is important to note that, by doing this, we are modeling a specific topology, meaning that on a global scale and across different countries, the topology of the network would change and, consequently, the infection prediction would also be different.

The pipeline is divided into three major modules: data retrieval, network generation and visualization.

Data retrieval

Three different sets of data were used in this work, corresponding to air, land and water routes.

Airborne transmission was modeled using the OpenFlights dataset, from which airport locations and flight routes were extracted. In this network, nodes correspond to airports mapped via GeoName cities, while edges are constructed based on the flight routes, with weights representing the flight frequency. Although this data is not modelling a real infection spread across regions, it effectively captures the potential pathways of aviation-mediated pathogen diffusion.

Land and water layers were also considered, approximating the generated network to a real-life epidemic scenario. Road and rail routes were considered for land transmission, while ferry routes represented the water transmission. **Data for these two additional layers were obtained from OpenStreetMap (OSM).**

When considering only air transmission routes, a diffusion mobility method was considered, meaning that migrating individuals permanently relocate to the destination node. Conversely, land and water commuting patterns were modeled using a recurrent mobility framework, which dictates that individuals return to the source node of origin.

Network generation

For the air-interdiction experiment 108 tests were executed. Across all iterations, the baseline parameters were held constant: 15 cities were designated for immunization,

vaccine efficacy was set at 85%, the population size per node was initialized with a seed of 2500 individuals, and the temporal horizon was fixed at 210 days.

As mentioned, 4 disease models were considered. For each, 5 different vaccination strategies were tested, including a control that did not implement any vaccination, a random vaccination that targeted random cities across Europe, targeted immunization applied to nodes exhibiting the highest degree centrality (local connectivity), as well as a vaccination applied to nodes exhibiting the highest betweenness centrality (bridging hubs).

By testing all these different strategies, the objective was to determine whether targeted vaccination, based on network metrics such as degree and betweenness, yields a significant suppression of disease spread compared to random immunization.

Furthermore, the vaccination coverage, i.e. the fraction of the population within the targeted nodes, was varied across three percentage thresholds (1%, 30% and 75%) to evaluate coverage-effect relation.

Axis	Values	Count
Disease model	SIR, SIS, SEIR, SQIR	4
Strategy	control, random, degree, betweenness	5
Coverage	1 %, 30%, 75 %	3 (control uses 1)

As mentioned, different infection models involve different parameters that were taken into account while building the simulations. The values used in this work are represented below, and were chosen based on existing literature.

	SIR	SIS	SEIR	SQIR
Transmission rate ($S \rightarrow I$): β	0.32	0.32	0.32	0.32
Recovery rate ($I \rightarrow R$ or $I \rightarrow S$): γ	0.12	0.12	0.12	0.12
Rate of progression ($E \rightarrow I$): σ	-	-	0.2	-
Rate of quarantine ($S \rightarrow Q$ or $I \rightarrow Q$, depending on the model): δ	-	-	-	0.15

Recovery rate of quarantined individuals ($Q \rightarrow R$): γ_Q	-	-	-	
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Disease types & parameters

The diseases we test, the compartmental model each uses, and the source for every parameter. Rates are per day; $\text{**rate} = 1/\text{duration**}$. β is not set directly — it is derived from R_0 and γ via $\beta = R_0 \cdot \gamma \cdot \langle k \rangle / \langle k^2 \rangle$.

Five disease types, defined by their dynamics. We give example diseases for each and take the parameter values from one representative exemplar (cited below).

Column 1	Type	Model	Example diseases	parameters from
1	Immunizing, acute	SIR	measles, rubella, mumps	measles
2	Latent + immunizing	SEIR	COVID-19, SARS, chickenpox	COVID-19
3	Lethal, isolation-controlled	SEIQR+D	Ebola, Marburg, pneumonic plague	Ebola
4	Endemic, no immunity	SIS	gonorrhea, chlamydia, common cold	gonorrhea
5	Recurrent, waning immunity	SEIRS	seasonal influenza, RSV, norovirus	seasonal influenza

Column 1	Type	γ	σ	κ	ω	μ	R_0
1	Immunizing · SIR	0.13 (~8 d)	—	0	0	≈ 0	12–18
2	Latent · SEIR	0.13 (~7–8 d)	0.20 (~5.1 d)	0	0	~ 0.01	2–3.3
3	Lethal · SEIQR+D	0.10 (~10 d)	0.09 (~11.4 d)	0.20 (~5 d)	0	0.71	1.5–2.5
4	Endemic · SIS	~ 0.01 – 0.05	—	0	∞	0	low (≥ 1)
5	Recurrent · SEIRS	0.21 (~4.8 d)	0.70 (~1.4 d)	0	~ 0.001	~ 0.001	1.3–1.8

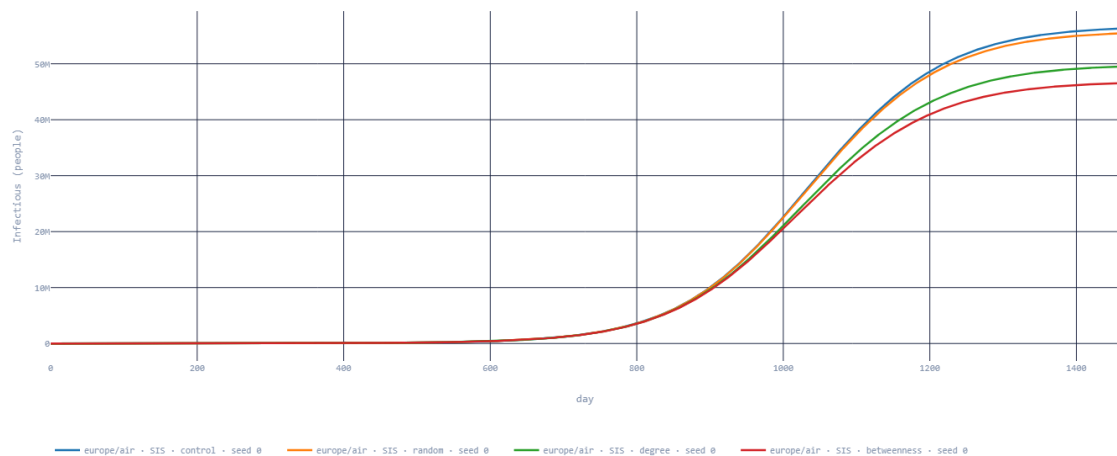
The only one we don't have (and dont worry mr lucas and mr matilde its just a SEIQR variation) is the lethal disease model. The sources are in /docs/paper-methodology/[disease-types.md](#) in github :)

Results

SIS model

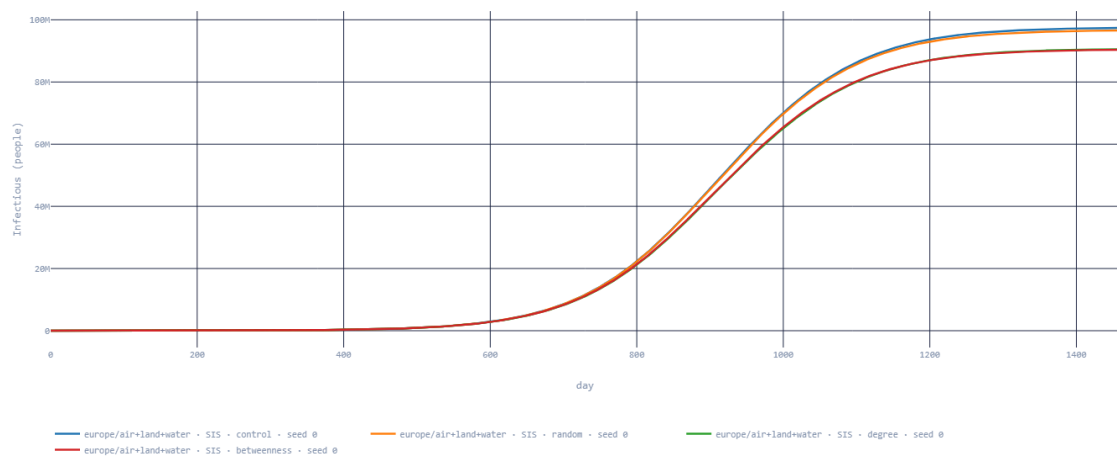
Infections modeled with the SIS model indicated a peak value of infections on day 1459, showing a logistic S-curve for all vaccination strategies tested. Compared to the control (no vaccination), random immunization showed the highest number on the peak infection day, while betweenness-based vaccination resulted in the least amount of infected individuals and the highest amount of vaccinated population.

Strategy	Peak infectious	Peak day	Vaccinated
control	56,293,171	1459	0
random	55,432,994	1459	3,071,887
degree	49,504,947	1459	24,012,258
betweenness	46,519,192	1459	34,671,063



The addition of the water and land layers to the networks resulted in an increase in the number of infected individuals overall, while maintaining the S-curve topology and the peak day. Moreover, a decrease was observed in vaccinated individuals. In this experiment, vaccination based on betweenness was also observed to be the strategy to result in less infected individuals and a higher number of vaccination, although, the difference between degree-based and betweenness-based immunization was noticed to be attenuated when using all three layers of transmission.

Strategy	Peak infectious	Peak day	Vaccinated
control	97,393,565	1459	0
random	96,585,968	1459	2,840,806
degree	90,540,322	1459	24,012,258
betweenness	90,360,405	1459	24,670,218

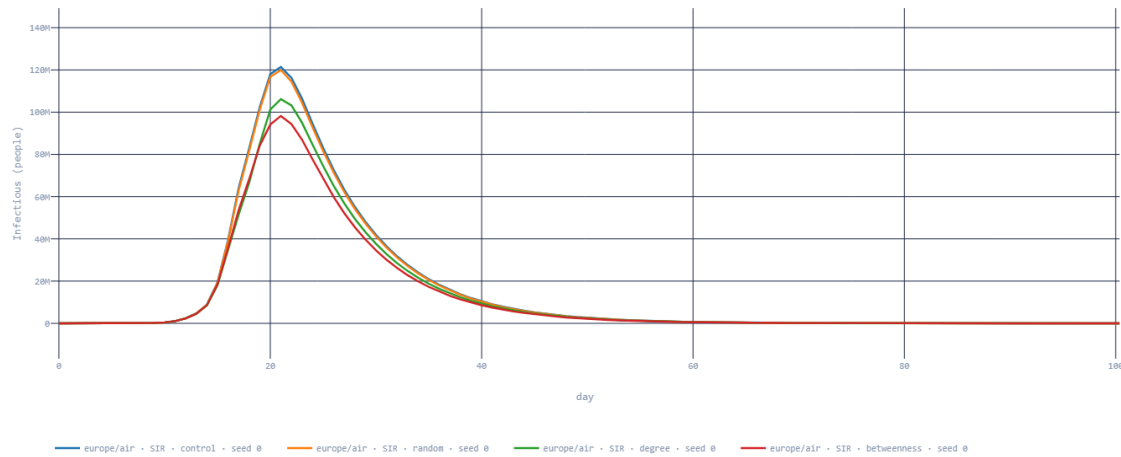


SIR model

In relation to the SIR model of infection, the infection evolution was analysed and compared across the different vaccination strategies tested (random, degree and betweenness based). From the plot generated, that shows the variation of the number of infected individuals across the simulation time, we can see that all strategies follow a similar infection evolution, with an infectious peak at day 21 and a gradual decrease in infected individuals that stays low until the end of the simulation.

In comparison with the control, all vaccination strategies show a decrease in the maximum number of people infected, with immunization in the 15 cities with higher betweenness centrality showing a less pronounced peak in relation to the other strategies. This vaccination strategy resulted in a lower number of total infections, as well as the highest number of people vaccinated, followed by degree and random vaccination.

Strategy	Peak infectious	Total infections	Vaccinated
control	121,401,738	198,617,855	0
random	119,951,185	195,545,969	3,071,887
degree	106,169,023	174,605,597	24,012,258
betweenness	98,202,188	163,946,793	34,671,063

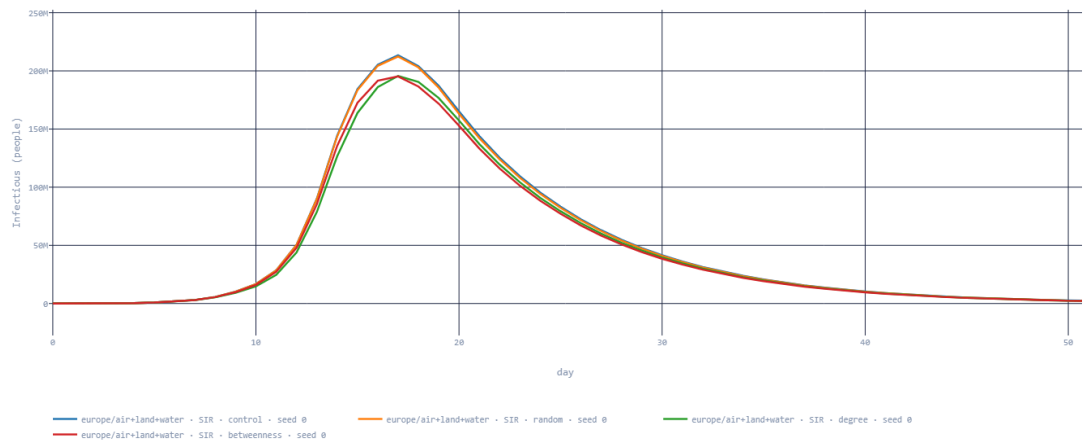


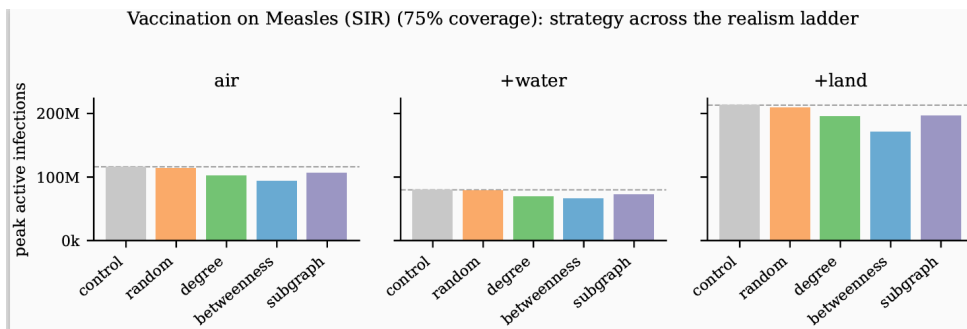
The addition of the different layer to the transmission network indicated the influence of these layers to the infection evolution and the consequences on the vaccination strategies. When adding the water layer, based on the ferry travel network, we see a decrease in the peak infections while, when adding a terrestrial layer, there is a significant increase in infections. Across all networks generated, betweenness based vaccinations shows a low infection number, in comparison with the other strategies tested, however, the difference between the number of infections between strategies is more pronounced in the multilayered network with air, water and land travels considered.

When adding the extra layers of transmission pathways, it was noticed that the infection evolution maintained its shape, however, the infectious peak for every condition happened earlier, at day 17. In comparison to transmission exclusively through the aviation network, the total number of infections with the additional land and water networks was much higher, while the total number of individuals vaccinated decreased. We can also see that there is less discrepancy between the number of infected individuals between degree and betweenness vaccination, in comparison to air transmission.

Similarly to the one-layer network, betweenness-based vaccination showed a lower number of total infections, followed by degree based and random vaccination.

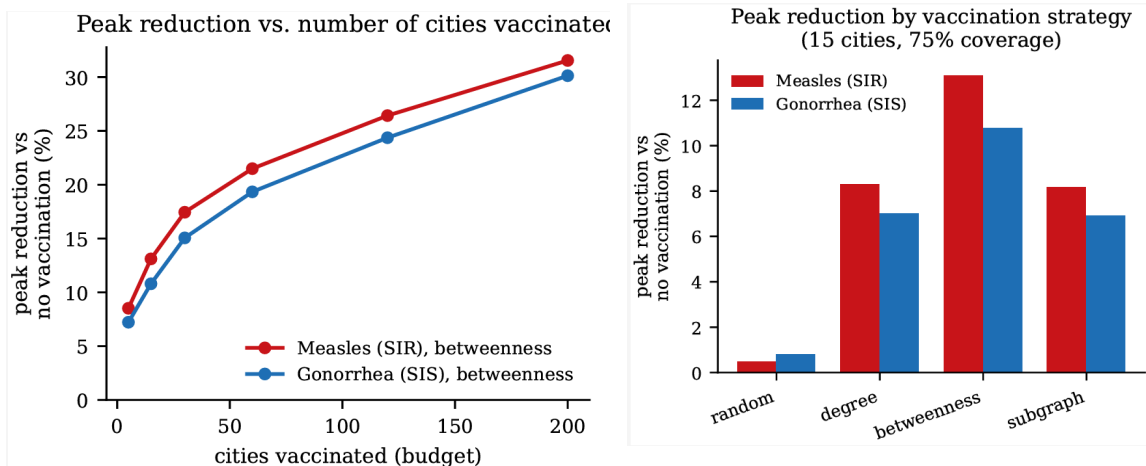
Strategy	Peak infectious	Peak day	Total infections	Vaccinated
control	213,423,737	17	341,512,518	0
random	212,373,137	17	338,671,712	2,840,806
degree	195,665,861	17	317,500,260	24,012,258
betweenness	195,245,083	17	316,842,301	24,670,218





As mentioned, the models constructed are comparative and not predictive, as such the infection evolution was found to change according to parameters that were fixed during the simulation. The runs were performed with a total of 15 cities vaccinated, however, it was observed that, for both infection models, the decrease in infected individuals tends to increase with the increase of cities vaccinated.

Furthermore, by comparing the reduction of the infected peak across vaccination strategies for both models, we can observe that the SIR model tends to show a higher reduction in relation to the SIS model for every vaccination strategy, except random.



- + **CONTROL NETWORK AND BETWEENESS**
- + **AIR NETWORK AND MULTILAYER (BETWENESS)**

Discussion

Although epidemic propagation can occur in many different ways, this study focused mainly on the transmission happening via air travel, via the European aviation network. Realistically, infection does not propagate only through air travel and even the extension of the propagation to a multilayer network with additional land and water travels cannot fully simulate the transmission of an infectious disease. Moreover, the present study aimed to study disease propagation in Europe, which, due to network topology, is expected to transmit a different transmission simulation than other regions or even globally, which is supported by literature.

It is also important to understand that in this simulation, certain parameters like the number of cities immunised and vaccine efficacy were fixed, which can have effects on the infection evolution, which can be observed on figure x, where a higher number of cities vaccinated leads to a higher reduction of infection, which would be expected, regardless of vaccination strategy or disease model, supporting the study as a comparative study and not a realistic prediction. Moreover a fixed coverage of 75% was used for the tests made, however, socioeconomic factors may change the way vaccination coverage is applied to a certain city or region.

As mentioned, SIR and SIS disease models were tested and evaluated, fixed with parameters supported by literature, representing a highly transmissible immunizing type and the slow endemic type with no immunity, respectively. Regarding the SIR model, the epidemic exhibits a unimodal, right-skewed incidence curve, characteristic of SIR models []. In this model, at each timestamp, each infected node infects each of its susceptible neighbors with probability β and recovers with rate γ []. This network-based transmission translates to the observed curve with the initial growth phase occurring because infected individuals have access to many susceptible neighbours, allowing transmission to propagate efficiently through the highly connected contact network. As the epidemic progresses, however, an increasing proportion of neighbouring nodes become infected and recover, reducing the number of susceptible nodes available. Consequently, a decrease in incidence is observed, this evolution is reinforced when vaccination strategies are added. By immunizing the 15 cities based on network centrality metrics (like degree and betweenness), it is possible to observe a decline in the infection peak, in relation to control, due to the closing of highly connected or bridging nodes from the network, that limit the transmission pathways between susceptible clusters which prevents the pathogen from spreading efficiently across the network. As a result, the epidemic peak is reduced and the outbreak declines more rapidly. This is more significantly observed when using betweenness based vaccination strategies, meaning that closing off airports that act as bridges in the network has a more limiting effect in aviation transmission than closing off airports with higher connectivity (hubs).

Conversely, the SIS model showed a S-shaped curve, with an increase in incidence that reaches an equilibrium with a high number of infected individuals for the simulation time. As referenced, SIS models are associated with diseases like gonorrhea, chlamydia or the common cold, with no immunity, which is consistent with the expected network propagation of the type of epidemic portrayed, with a low relative transmission and recovery rate. Unlike the SIR model, where susceptible individuals become infected and then recover, the SIS model does not exhibit a permanent depletion of susceptible individuals, meaning that, after the initial period of increase of infected individuals, in which infected nodes transmit the disease to their neighbors, the epidemic reaches an endemic equilibrium rather than declining to zero. From a network perspective, transmission persists because recovered nodes continuously re-enter the pool of susceptible individuals, however, eventually, the number of new infections balances the number of recovered individuals, reaching the equilibrium observed. Similarly to the SIR model,

vaccination based on degree and betweenness demonstrated more efficiency in reducing the peak of infections, reducing the equilibrium, demonstrating how immunizing highly connected or bridging nodes disrupts transmission pathways and reduces the effective connectivity of the network. Consequently, infected individuals have fewer susceptible neighbours available to infect, decreasing the rate of new infections until it balances the recovery rate at a lower equilibrium level.

For both epidemic models, simulating the transmission on a multilayered network resulted in the same effect, where a higher number of infected individuals was observed, increasing the peaks obtained, and vaccination strategies resulted in an overall lower number of vaccinated individuals. When adding the additional layers, the network size grows significantly, resulting in a more connected network, which translates to a higher number of transmission pathways, which can explain the growth in incidence for these networks. The decrease in vaccinated individuals can be explained by the fact that adding additional layers increases the redundancy and increases the existence of alternative transmission routes between the same set of nodes, meaning that the multilayer connectivity reduces the relative importance of individual nodes in interrupting transmission pathways. As a result, structured based vaccination strategies (degree and betweenness) become less efficient, because there are more alternative pathways the infection can follow.

It was also noticed that adding a water layer decreased the active infection peak across all conditions tested, in relation to the air layer, while the additional land layer caused a massive increase in infection peak.

Overall, betweenness centrality was the metric that showed more efficiency for immunization across the two models tested for one-layered and multi-layered networks. This metric identifies vertices with the shortest paths between all node pairs, effectively capturing nodes that mediate inter- and intra-community connectivity. Because the epidemic process is modeled on networks, transmission is strongly correlated with path accessibility. Immunizing high-betweenness nodes reduces the number of available shortest infection pathways, thereby increasing effective path lengths, decreasing global network efficiency, and fragmenting the largest connected component.

In multilayer networks, this effect is amplified because shortest-path structure is distributed across multiple interaction channels. Nodes with high multilayer betweenness often function as interlayer conduits, so their removal simultaneously disrupts multiple diffusion channels, leading to a disproportionate reduction in reachable node sets and epidemic transmissibility compared to degree-based or random immunization strategies.

Furthermore, the more significant reduction in infection observed in the SIR model is attributable to the presence of the recovered state, which induces cumulative depletion of the susceptible population and irreversible disruption of transmission pathways. In contrast, the SIS allows reinfection, which sustains epidemic circulation and attenuates the long-term effectiveness of immunization strategies.

To more accurately study the epidemic evolution using networks, more models, like SEIR and SQIR can be used, as well as different parameters and layers, that could potentially lead to a more informed study and higher level of comparisons. With the app build during this study, this can be done, furthering the potential of studying epidemic progression using travel networks.

Conclusion

This study investigated epidemic propagation across the European aviation network using single-layer (air) and multilayer (air+land+water) representations under SIR and SIS model dynamics. The results show that network structure and layer composition significantly influence both epidemic intensity and the effectiveness of immunization strategies. Across both epidemic models, multilayer networks generally increased transmission potential due to higher connectivity and the presence of additional pathways between the same set of nodes. Vaccination strategies based on network centrality consistently reduced infection incidence, with betweenness centrality emerging as the most effective vaccination strategy, reflecting the effectiveness of targeting cities that mediate shortest-path connectivity and act as bridges within the networks. Differences between epidemic models further highlight the role of disease dynamics in shaping intervention outcomes, with the SIR model producing a single-peaked epidemic curve with strong sensitivity to immunization, while the SIS model exhibited endemic equilibrium behavior, with reinfection sustaining long-term circulation and reducing the overall impact of vaccination strategies.

Overall, the results demonstrate that both network topology and disease models determine epidemic outcomes and influence the effectiveness of vaccination strategies.

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